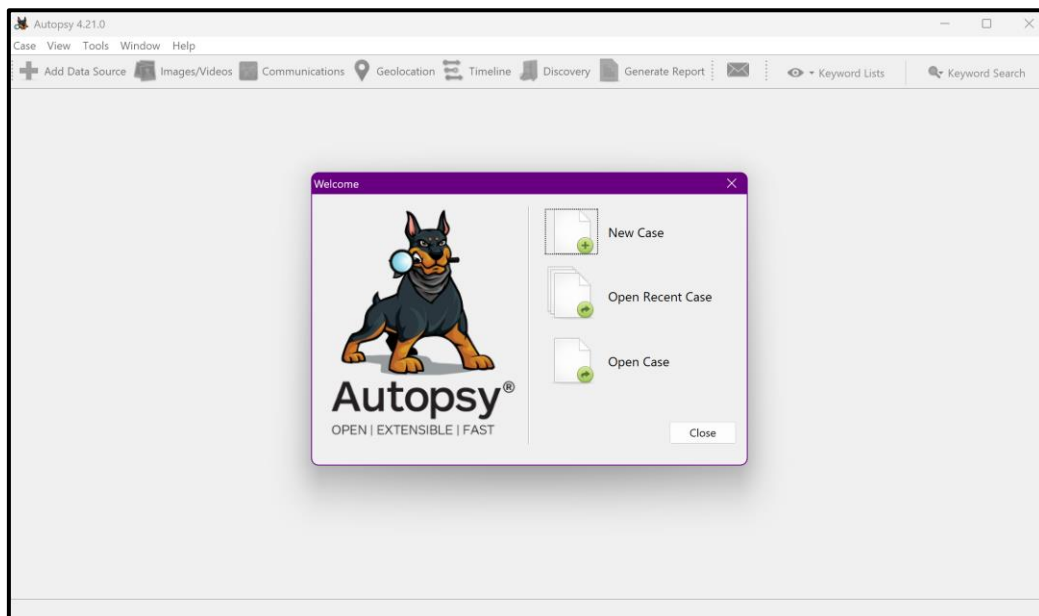


Lab 2 DFIR : Autopsy Analysis

-Eshan Shende : K027



USE OF AUTOPSY

Autopsy is a robust, open-source digital forensics platform designed to simplify the process of analyzing and extracting evidence from electronic devices. Widely used by law enforcement agencies, cybersecurity professionals, and forensic investigators, it provides a comprehensive suite of tools to investigate disk images, devices, and file systems through an intuitive graphical user interface (GUI).

Developed by Basis Technology, Autopsy acts as the GUI for The Sleuth Kit (TSK), a powerful set of command-line forensic tools, making advanced forensic analysis more accessible. It supports a wide range of file systems (e.g., NTFS, FAT, EXT), disk image formats, and forensic techniques such as artifact analysis, keyword searching, timeline reconstruction, hash matching, and metadata extraction. Autopsy also integrates modules to analyze common digital artifacts like web browsing history, email, geolocation data, and chat logs, providing deep insights into user activity. Its extensibility through custom plugins, cross-platform compatibility, and ability to support multi-user collaboration make it an indispensable tool for modern digital investigations.

UNDERSTANDING FEATURES

Features of Autopsy

1. **File Analysis and Metadata Extraction**

Autopsy allows investigators to analyze files and extract metadata, such as timestamps (created, modified, accessed), size, type, and ownership. This helps trace a file's history and establish its relevance to an investigation.

2. **Data Carving**

This feature recovers deleted or fragmented files by identifying unique file signatures, even if the file system metadata is missing. It is particularly useful for retrieving evidence from unallocated or damaged disk space.

3. **Timeline Analysis**

Autopsy builds a chronological timeline of events by correlating file activities, such as creation, modification, access, or deletion. This visual representation helps investigators pinpoint when key actions occurred.

4. **Web and Browser Analysis**

Recovers browser history, cookies, and cached files from web browsers. This feature is critical for tracking a suspect's online activity, identifying visited websites, and uncovering hidden or anonymized browsing.

5. **Keyword Search**

Investigators can search for specific terms or phrases within files, logs, and unallocated space. This feature also supports advanced search techniques, like regular expressions, for targeted results.

Importance of Autopsy

1. **Open-Source and Cost-Effective**

Being open-source, Autopsy is free and accessible to organizations or individuals with limited budgets, making it a practical choice for academic, small-scale, and large-scale investigations.

2. **Court-Admissible Evidence Handling**

Autopsy ensures that evidence is processed and preserved in a forensically sound manner, maintaining its integrity and admissibility in court proceedings.

3. **Comprehensive Toolset**

With features like file analysis, timeline building, and browser history extraction, Autopsy combines multiple forensic capabilities into a single platform, streamlining investigations.

4. **Extensibility Through Modules**

Autopsy's modular architecture allows integration with plugins and third-party tools, enabling users to customize it for specific investigative needs, such as malware detection or encrypted file analysis.

Uses of Autopsy

1. Criminal Investigations

Autopsy is widely used by law enforcement agencies to uncover evidence of crimes like fraud, harassment, or exploitation by recovering deleted files or analyzing online activity.

2. Corporate Incident Response

Organizations use Autopsy to investigate data breaches, insider threats, or compliance violations. It helps track unauthorized access or misuse of sensitive data.

3. Cybersecurity and Malware Analysis

Autopsy can analyze system logs, browser data, and disk activity to detect malware, track its origin, and understand its impact on a compromised system.

4. Mobile Forensics

With additional modules, Autopsy can extract and analyze data from mobile devices, including call logs, messages, and app data, making it suitable for modern investigations.

Functions of Autopsy

1. Data Recovery

Autopsy excels at recovering deleted, hidden, or lost data, including files that have been fragmented or partially overwritten, ensuring that critical evidence is not overlooked.

2. Artifact Analysis

It allows investigators to focus on digital artifacts, such as registries, browser history, or email logs, to reconstruct user actions and identify potentially malicious behavior.

3. Case Management

Autopsy organizes evidence systematically, providing tagging features, bookmarking, and detailed reporting options to help investigators maintain a clear workflow.

4. Automation of Forensic Tasks

Automated processes, such as hash filtering, keyword searches, and timeline building, save time and improve efficiency during investigations by handling repetitive tasks.

5. Evidence Reporting

Autopsy generates detailed, customizable reports in formats like HTML and CSV, enabling investigators to present findings in a clear and professional manner suitable for court or stakeholders.

ACTIVITIES PERFORMED

Understanding the various elements of the autopsy application:

Add Data Source (Green Plus Icon)

Function: Allows users to import data sources, such as disk images, logical files, or folders, for forensic analysis.

Use: Serves as the starting point for investigations by enabling the import of evidence.

Images/Videos (Photo and Film Strip Icon)

Function: Provides tools for browsing, analyzing, and tagging image and video files.

Use: Helps identify relevant visual content, including photos, screenshots, or video recordings.

Communications (Envelope Icon)

Function: Focuses on analyzing communication data, such as emails, chat messages, and call logs.

Use: Critical for investigating communication patterns, identifying persons of interest, or extracting evidence from conversations.

Geolocation (Pin Icon)

Function: Displays geographical data, such as GPS coordinates found in media files or applications.

Use: Useful for mapping locations or tracking movements relevant to the investigation.

Timeline (Bar Chart Icon)

Function: Generates a timeline of events based on timestamps, like file creation, modification, or access times.

Use: Aids in identifying sequences of events or pinpointing timeframes of interest.

Discovery (Document Icon)

Function: Assists in finding specific files, such as documents or images, within the data.

Use: Automates searches and categorization, speeding up the process of locating critical evidence.

Generate Report (Document with Pie Chart Icon)

Function: Enables the creation of reports summarizing findings in various formats, such as HTML or PDF.

Use: Useful for documenting results for court presentations, collaboration, or archiving.

Keyword Lists (Eye Icon with Dropdown)

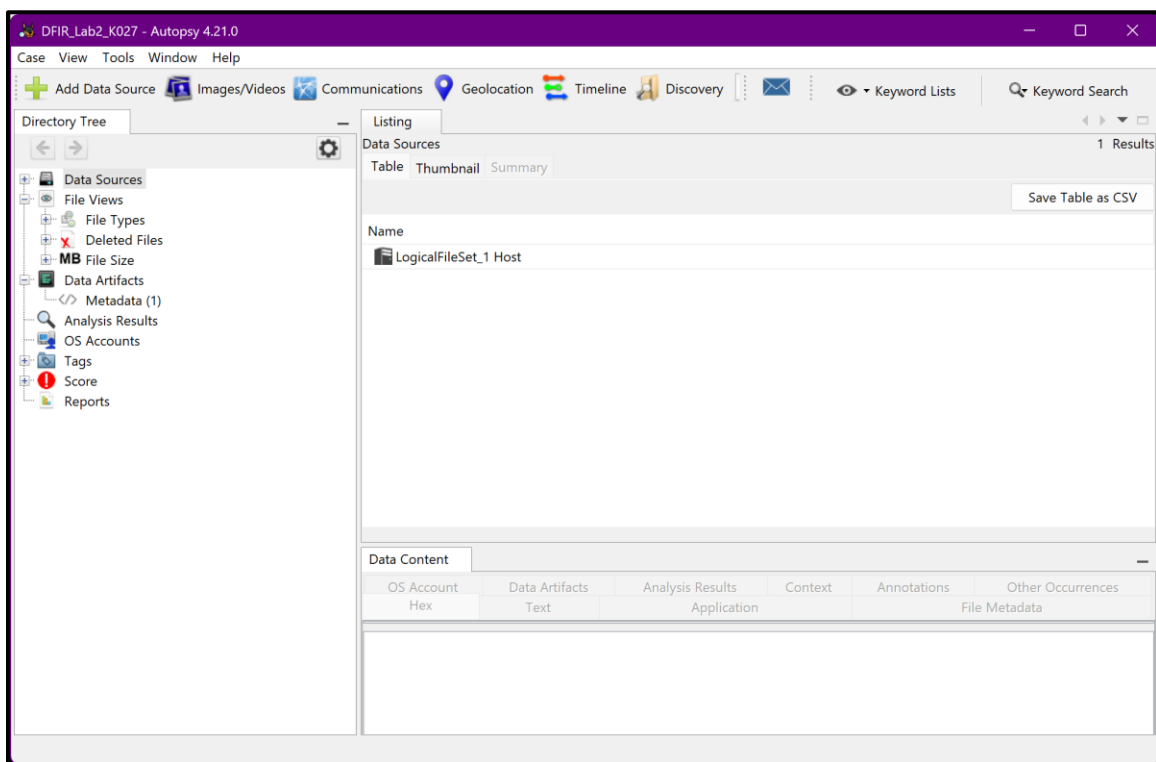
Function: Manages and applies predefined or custom keyword lists for searching across the evidence.

Use: Efficiently identifies text data, like emails or documents containing specific terms.

Keyword Search (Magnifying Glass Icon)

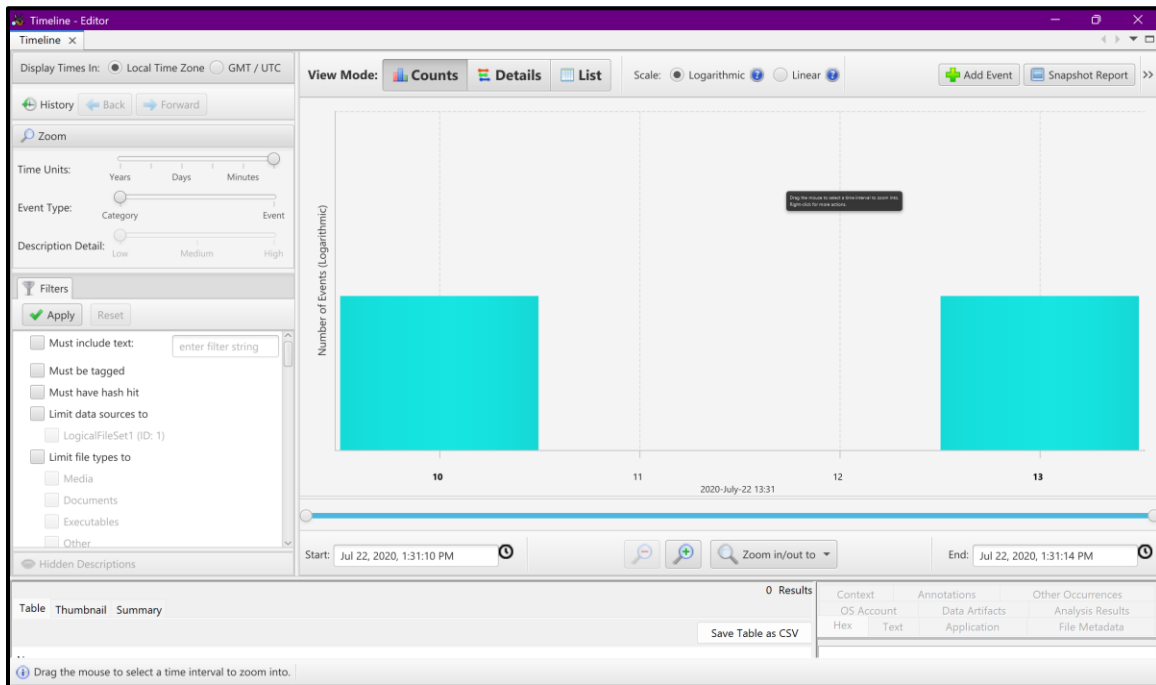
Function: Provides a manual search interface for entering and locating specific keywords.

Use: Allows investigators to pinpoint specific information without relying on predefined lists.



Exploring various features via hands on method by analyzing a pdf document to engineer the digital evidences of the materials under analysis and create conclusions by observing changes over time and factual analysis

The image depicts a time analysis of a PDF document .



The image depicts the image [photograph] analysis of the PDF document.

